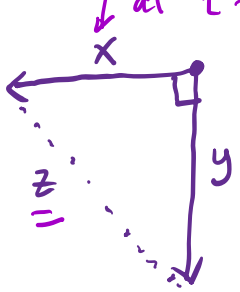
Problem 2. WORKED IN CLASS

Two cars leave the same point at the same time. One drives **south** at 60 mi/h, the other **west** at 25 mi/h. How fast is the distance between them changing **two hours** later?

1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. t
5. Sub + Solve

(1)



at $t=2$ $x=50$

$y = 2 \cdot 60 = 120$

(2) want $\frac{dz}{dt}$ when $t=2$

know $\frac{dx}{dt} = 25$, $\frac{dy}{dt} = 60$

(3) $z^2 = x^2 + y^2$

(4) $\frac{d}{dt}(z)^2 = \frac{d}{dt}x^2 + \frac{d}{dt}y^2$

$2z\frac{dz}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$

(5) first solve to get z

$z^2 = (50)^2 + (120)^2$

$z^2 = 2500 + 14,400$

$z^2 = 16,900$

$z = 130$

$130\frac{dz}{dt} = 50 \cdot 25 + 120 \cdot 60$

$130\frac{dz}{dt} = 8,450$

$\frac{dz}{dt} = \frac{8,450}{130} = 65 \frac{\text{miles}}{\text{hour}}$